

# Java Foundation: 25-Problem Roadmap

Objective: Achieve mastery in logic, from fundamental syntax to nested loops, within a 5-day period.

## Day 1: Variables & Fundamental Arithmetic

1. Problem 1: Compute Simple Interest (SI) using Principal (P), Rate (R), and Time (T).  
Formula:  $SI = (P \times R \times T) / 100$ .
2. Problem 2: Determine a circle's area and circumference based on radius (r).  
Formulas:  $Area = \pi r^2$ ,  $Circumference = 2\pi r$ .
3. Problem 3: Exchange the values of two variables without employing a third temporary variable.
4. Problem 4: Output the ASCII value for any character provided by the user.
5. Problem 5: Input three integers and calculate their mean average.

## Day 2: Conditional Logic & Decision Making

1. Problem 6: Identify whether a given integer is Even or Odd.
2. Problem 7: Utilize nested if-else structures to find the maximum among three numbers.
3. Problem 8: Classify an input character as either a vowel or a consonant.
4. Problem 9: Build a basic calculator using switch-case for operations: +, -, \*, or /.
5. Problem 10: Evaluate if a number is positive, negative, or zero.

## Day 3: Standard Iteration & Loops

1. Problem 11: Calculate the total sum of the first N natural numbers.
2. Problem 12: Generate the multiplication table for any number N up to 10.
3. Problem 13: Invert an integer's digits (e.g., transform 1234 into 4321).
4. Problem 14: Determine the total number of digits present in an integer.
5. Problem 15: Compute X raised to the power of Y manually, avoiding Math.pow().

## Day 4: Complex Loop Logic

1. Problem 16: Verify if a specific number is Prime.
2. Problem 17: Display the Fibonacci sequence up to the Nth term.
3. Problem 18: Check for palindrome numbers (e.g., 121 remains the same when reversed).
4. Problem 19: Confirm if a number is an Armstrong number (e.g.,  $153 = 1^3 + 5^3 + 3^3$ ).
5. Problem 20: Identify and list all factors of a provided integer.

## Day 5: Nested Loops, Patterns & Advanced Math

1. Problem 21: Construct a right-angled triangle using asterisk (\*) symbols.
2. Problem 22: Create an N x N square grid using stars.
3. Problem 23: Render an inverted triangle pattern of stars.
4. Problem 24: Generate a numerical pyramid (e.g., Row 3 displays "1 2 3").
5. Problem 25: Solve for the sum of factorials for every digit within a number.